roam: Remote Objects with Active-binding Magic

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Including data in a package without `roam`

Package pkg Writer

```r
# data-raw/bee.R
library(readr)
bee <- read_csv(<bee.csv>)
usethis::use_data(bee)
```

User

```r
library(pkg)
bee
```

```
# A tibble: 1,222 x 10
   year months  state colony_n
  <dbl> <chr>   <chr>    <dbl>
1  2015 Januar~ Alab~     7000
2  2015 Januar~ Ariz~    35000
# i 1,220 more rows
# i 6 more variables:
#   colony_max <dbl>,
#   colony_lost <dbl>,
#   colony_lost_pct <dbl>,
#   colony_added <dbl>,
#   colony_reno <dbl>, ...
```

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library(readr)
bee <- read_csv(<bee.csv>)
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```

- Data included in package
- CRAN size limit 5 MB
- The data are updated only when the package is updated

User

```r
library(pkg)
bee
```

```
# A tibble: 1,222 x 10
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```

Including data in a package with `roam`

Package pkg Writer

```
obtainer <- function(version)
{
  readr::read_csv(<bee.csv>)
}

bee <- roam::new_roam(
  'pkg', 'bee', obtainer)

.onLoad <- function(libname,
                    pkgname)
{
  roam::roam_activate(bee)
}
```

User

```
library(pkg)
options(roam.autodownload = TRUE)
bee
```

```
# A tibble: 1,222 x 10
   year months  state colony_n
  <dbl> <chr>   <chr>    <dbl>
1  2015 Januar~ Alab~     7000
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Including data in a package with `roam`

Package pkg Writer

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obtainer <- function(version)
{
  readr::read_csv(<bee.csv>)
}

bee <- roam::new_roam(
  'pkg', 'bee', obtainer)

onLoad <- function(libname
```

- Data hosted remotely
- No size limit
- Updating data without updating package

User

```
library(pkg)
options(roam.autodownload = TRUE)
bee
```

```
# A tibble: 1,222 x 10
   year months  state colony_n
  <dbl> <chr>   <chr>    <dbl>
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#   colony_reno <dbl>, ...
```

Versioning with `roam`

Package pkg Writer

```
obtainer <- function(version) {
  if(version == 'latest')
    version <- '0.1.0'
  # Specify the version
  roam::roam_set_version(version)
  readr::read_csv(<bee-v.csv>)
}

bee <- roam::new_roam(
  'pkg', 'bee', obtainer)

.onLoad <- function(libname,
                    pkgname)
  roam::roam_activate(bee)
```

User

```
library(pkg)
library(roam)

# Update to the latest version
roam_update(bee)

# Install a specific version
roam_install(bee, version = '0.1.0')

# Delete local cache
roam_delete(bee)

# Find currently installed version
roam_version('pkg', 'bee')
```

More information

Roadmap: Cache multiple versions

```r
library(pkg)
library(roam)

bee %v% "0.1.0"
bee %v% "0.1.1"
```

More information

Roadmap: Cache multiple versions

```r
library(pkg)
library(roam)

bee %v% "0.1.0"
bee %v% "0.1.1"
```

Slides: yangzhuoranyang.com/talk/user-2024
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